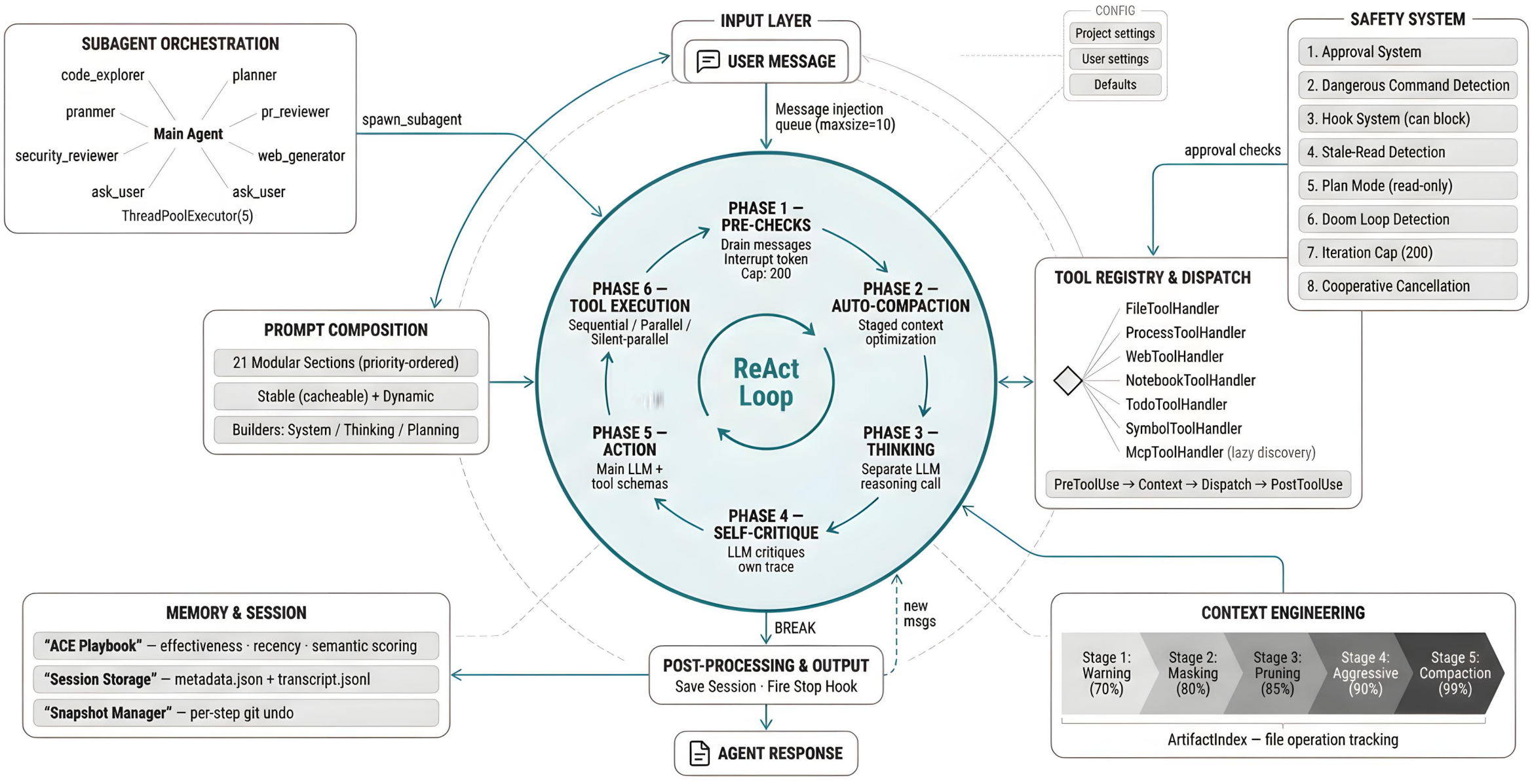
SUBAGENT ORCHESTRATION
code_explorer
planner
pranmer
pr_reviewer
Main Agent
security_reviewer
web_generator
ask_user
ask_user
ThreadPoolExecutor(5)
spawn_subagent
INPUT LAYER
USER MESSAGE
Message injection queue (maxsize=10)
CONFIG
Project settings
User settings
Defaults
SAFETY SYSTEM
1. Approval System
2. Dangerous Command Detection
3. Hook System (can block)
4. Stale-Read Detection
5. Plan Mode (read-only)
6. Doom Loop Detection
7. Iteration Cap (200)
8. Cooperative Cancellation
approval checks
PHASE 1 — PRE-CHECKS
Drain messages
Interrupt token
Cap: 200
PHASE 2 — AUTO-COMPACTION
Staged context optimization
PHASE 3 — THINKING
Separate LLM reasoning call
PHASE 4 — SELF-CRITIQUE
LLM critiques own trace
PHASE 5 — ACTION
Main LLM + tool schemas
PHASE 6 — TOOL EXECUTION
Sequential / Parallel / Silent-parallel
ReAct Loop
PROMPT COMPOSITION
21 Modular Sections (priority-ordered)
Stable (cacheable) + Dynamic
Builders: System / Thinking / Planning
TOOL REGISTRY & DISPATCH
FileToolHandler
ProcessToolHandler
WebToolHandler
NotebookToolHandler
TodoToolHandler
SymbolToolHandler
McpToolHandler (lazy discovery)
PreToolUse → Context → Dispatch → PostToolUse
MEMORY & SESSION
"ACE Playbook" — effectiveness · recency · semantic scoring
"Session Storage" — metadata.json + transcript.jsonl
"Snapshot Manager" — per-step git undo
BREAK
new msgs
POST-PROCESSING & OUTPUT
Save Session · Fire Stop Hook
AGENT RESPONSE
CONTEXT ENGINEERING
Stage 1: Warning (70%)
Stage 2: Masking (80%)
Stage 3: Pruning (85%)
Stage 4: Aggressive (90%)
Stage 5: Compaction (99%)
ArtifactIndex — file operation tracking